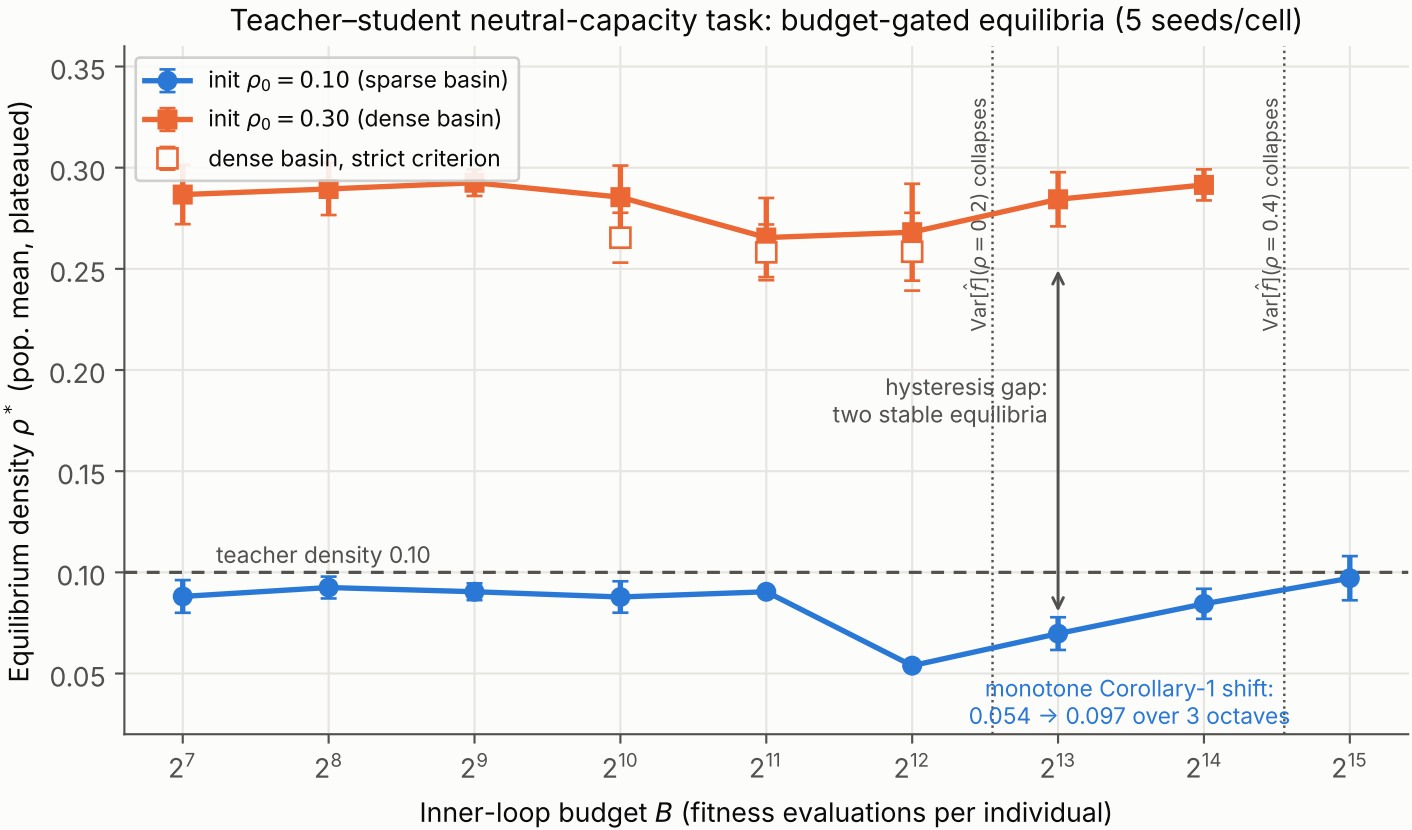
Teacher–student neutral-capacity task: budget-gated equilibria (5 seeds/cell)
init $\rho_0 = 0.10$ (sparse basin)
init $\rho_0 = 0.30$ (dense basin)
dense basin, strict criterion
Equilibrium density ρ^* (pop. mean, plateaued)
0.35
0.30
0.25
0.20
0.15
0.10
0.05
teacher density 0.10
$\mathrm{Var}[\hat{f}](\rho = 0.2)$ collapses
$\mathrm{Var}[\hat{f}](\rho = 0.4)$ collapses
hysteresis gap:
two stable equilibria
monotone Corollary-1 shift:
0.054 → 0.097 over 3 octaves
2^7
2^8
2^9
2^{10}
2^{11}
2^{12}
2^{13}
2^{14}
2^{15}
Inner-loop budget B (fitness evaluations per individual)